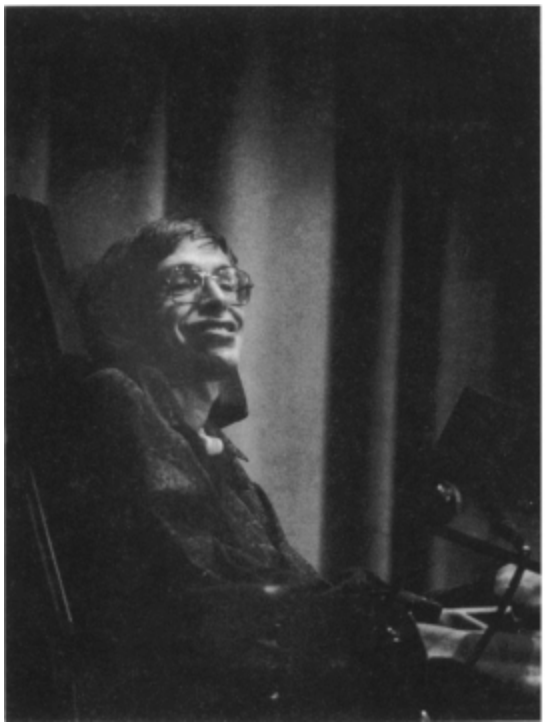
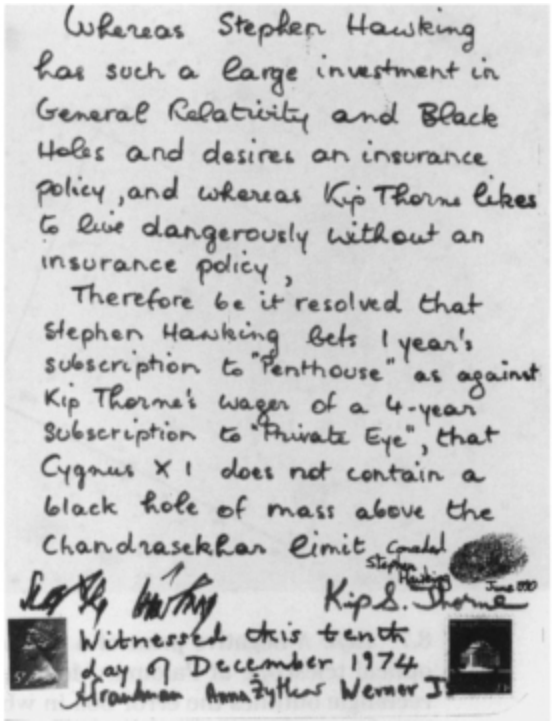
in June 1990, while I was in Moscow working on research with Soviet colleagues, Stephen and an entourage of family, nurses, and friends broke into my office at Caltech, found the framed bet, and wrote a concessionary note on it with validation by Stephen's thumbprint.





Right: **The bet between Stephen Hawking and me as to whether Cygnus X-1 is a black hole.** ***Left:*** **Hawking lecturing at the University of Southern California in June 1990, just two hours before breaking into my office and signing off on our bet.** [Hawking photo courtesy Irene Fertik, University of Southern California.]

The evidence that Cyg X-l contains a black hole is of just the sort that Zel'dovich and Novikov envisioned when they proposed the method of search: Cyg X-l is a binary made of an optically bright and X-ray-dark star orbiting around an X-ray-bright and optically dark companion, and the companion has been weighed to make sure it is too heavy to be a neutron star and thus is probably a black hole.

The evidence that this is the nature of eyg X-1 was not developed easily. It required a cooperative, massive, worldwide effort carried out in the 1960s and 1970s by hundreds of experimental physicists, theoretical astrophysicists, and observational astronomers.

The experimental physicists were people like Herbert Friedman, Stuart Bowyer, Edward Byram, and Talbot Chubb, who discovered eyg X-1 in a rocket flight in 1964; Harvey Tananbaum, Edwin Kellog, Herbert Gursky,